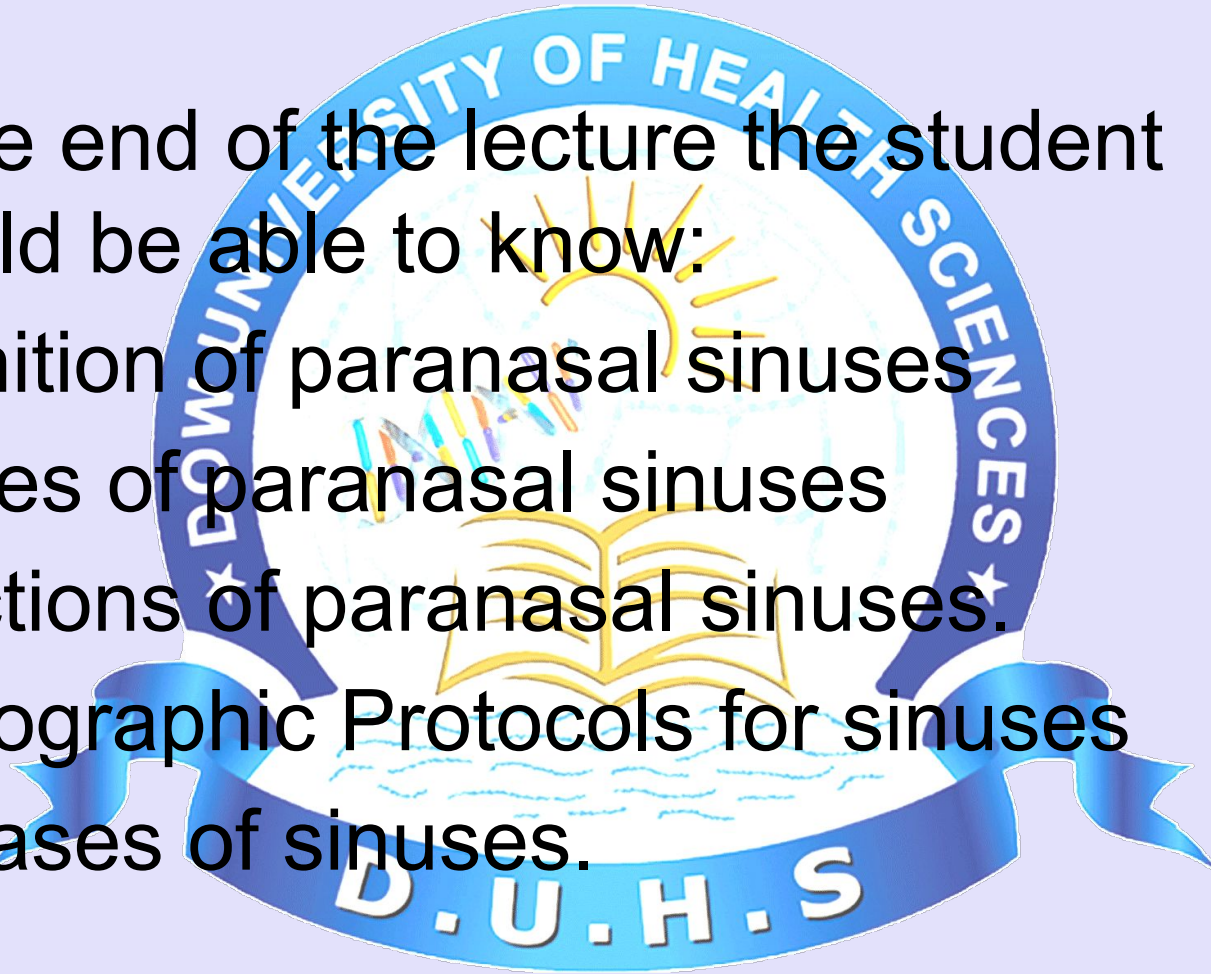


Paranasal sinuses



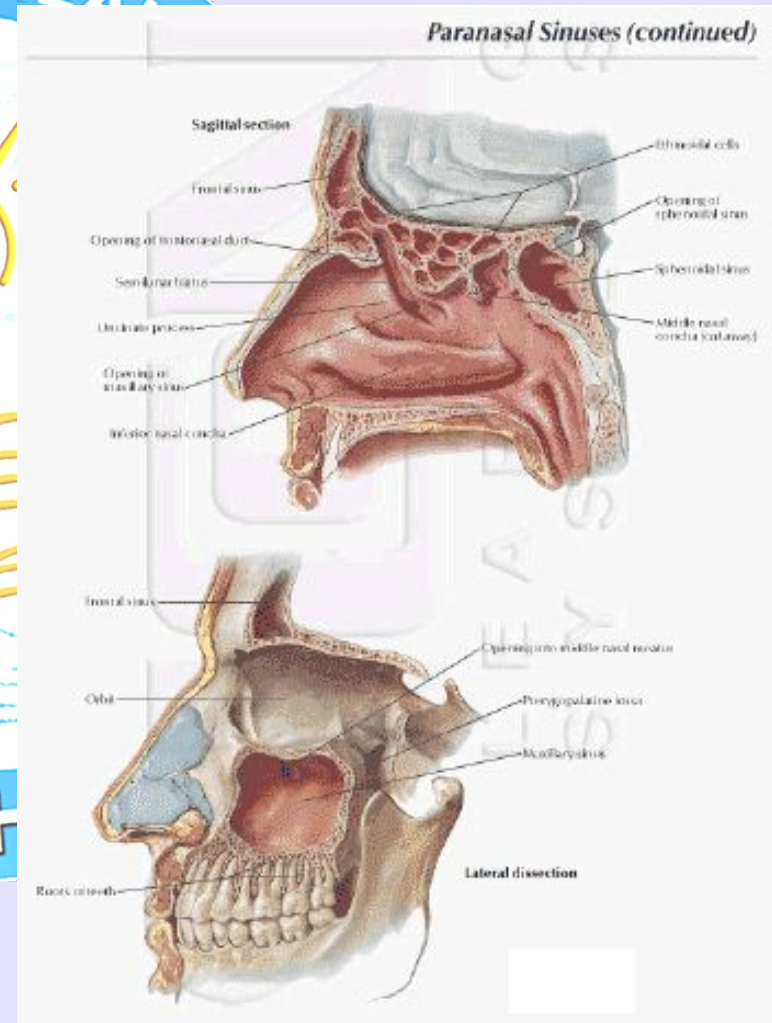
Learning objectives

- At the end of the lecture the student should be able to know:
- Definition of paranasal sinuses
- Names of paranasal sinuses
- Functions of paranasal sinuses.
- Radiographic Protocols for sinuses
- Diseases of sinuses.

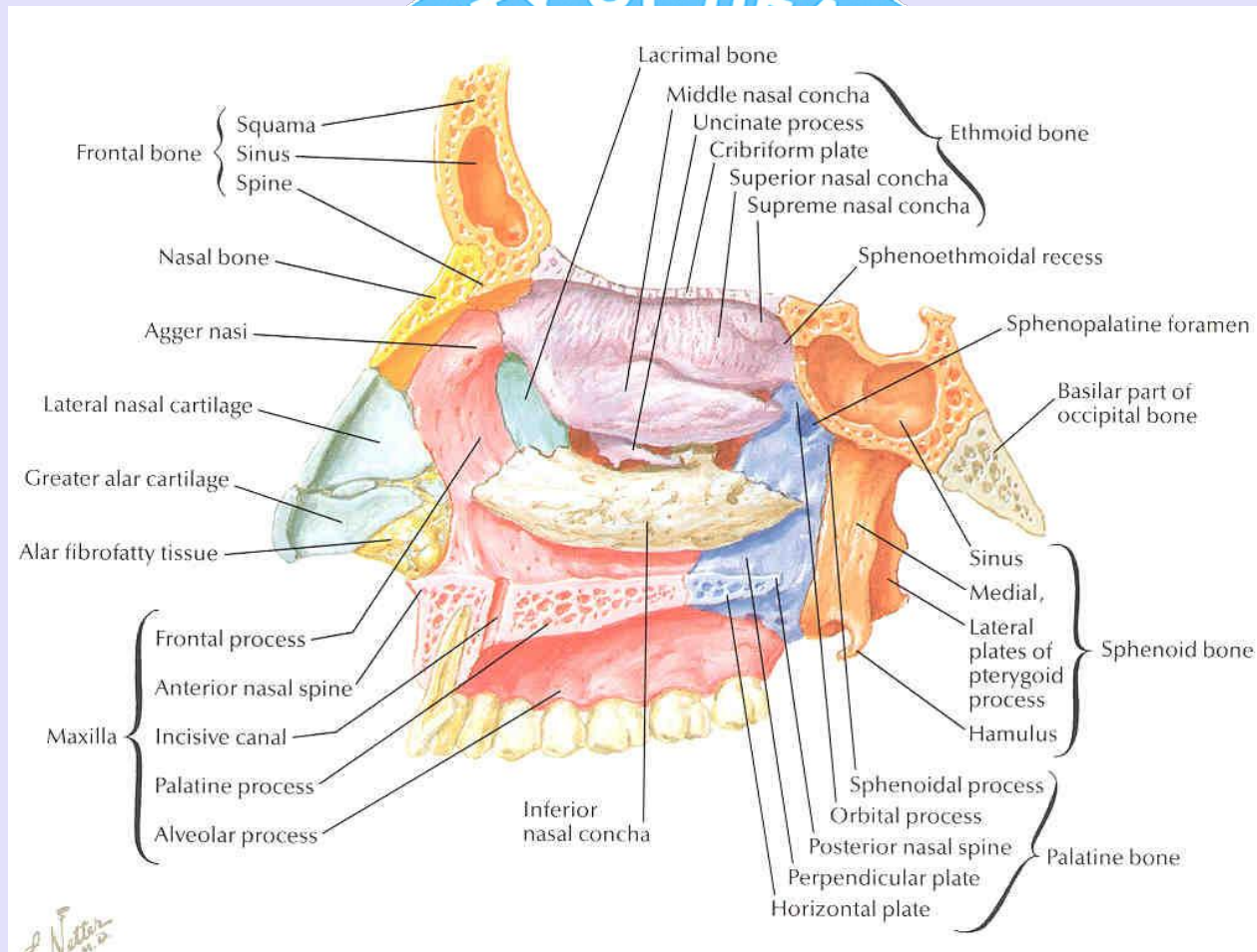


Paranasal sinuses

- **Paranasal sinuses** are air-filled spaces, communicating with the nasal cavity, within the bones of the skull and face.
- Divided into subgroups that are named according to the bones within which the sinuses lie

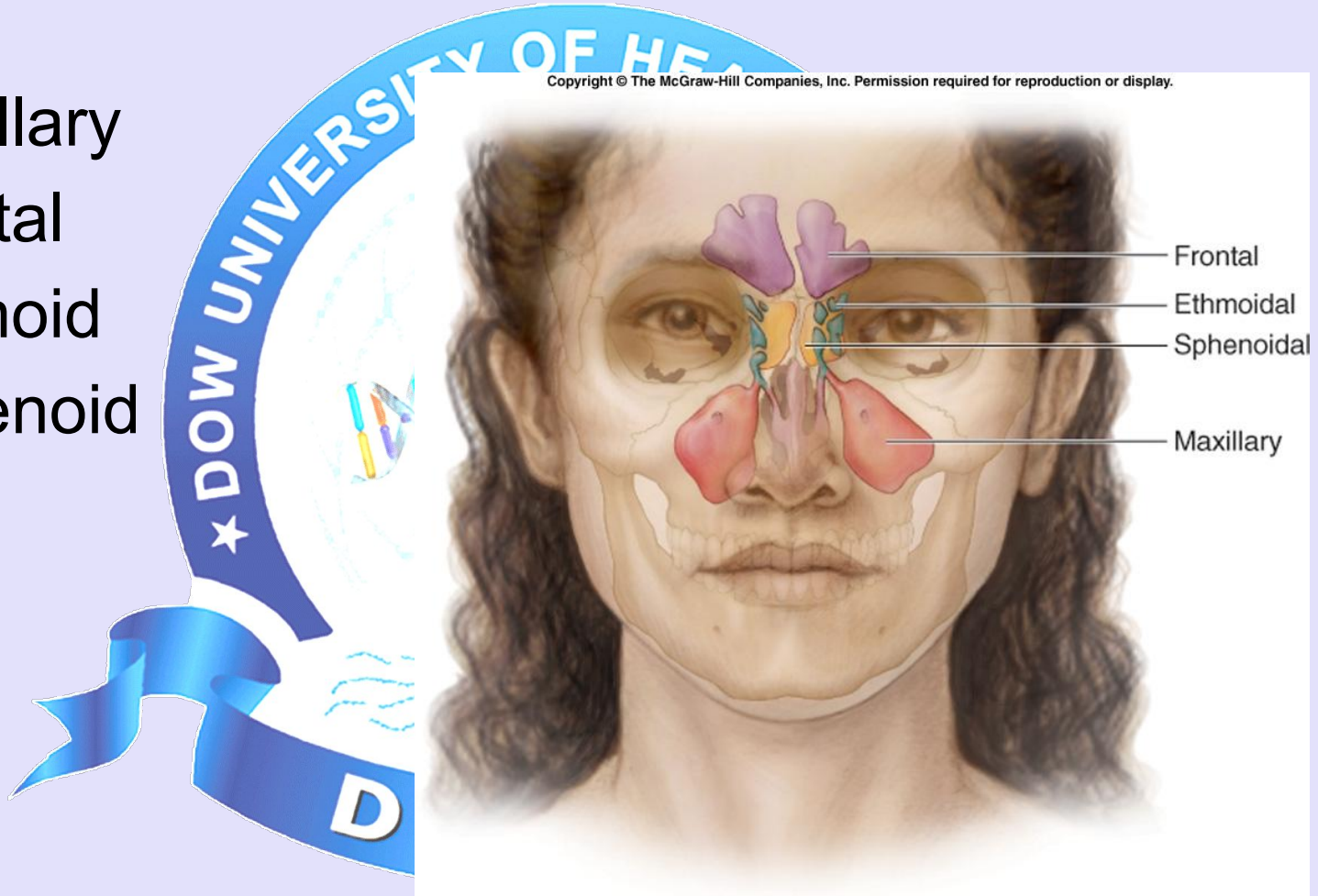


Bony structure of nose



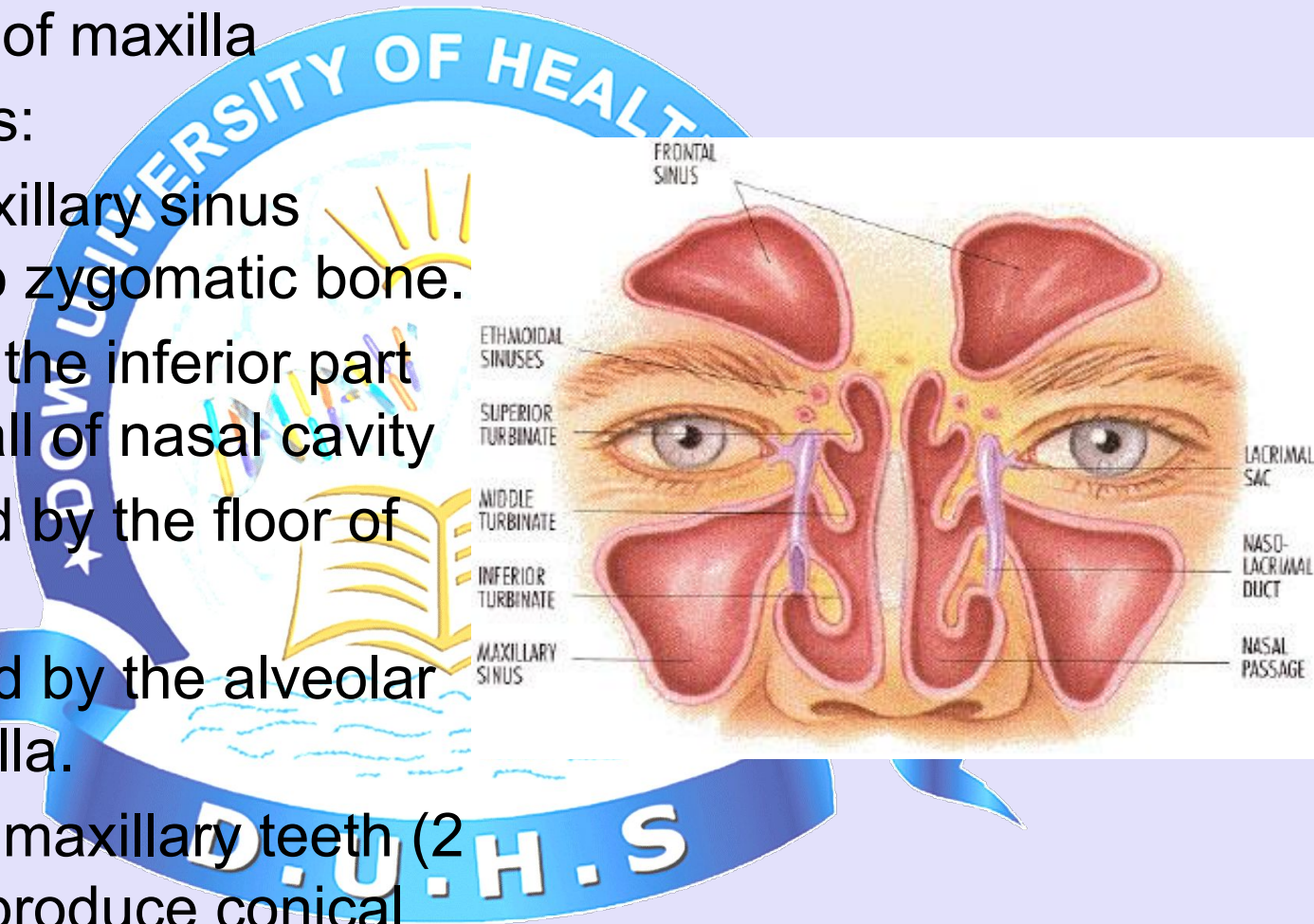
Names of sinuses

- Maxillary
- Frontal
- Ethmoid
- Sphenoid



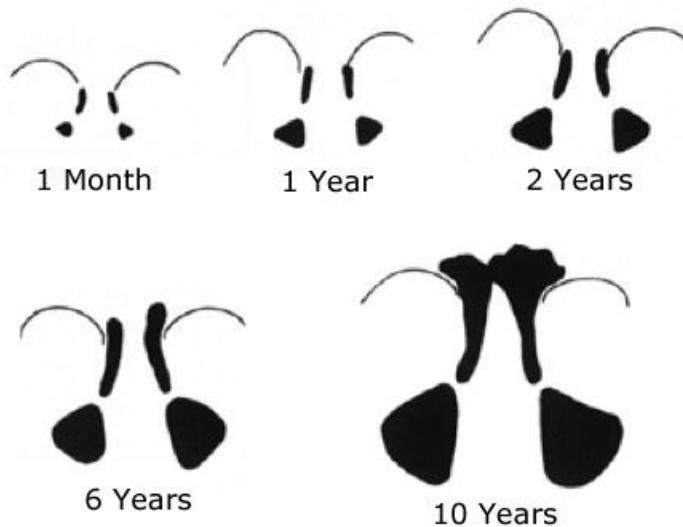
Maxillary sinus

- Present at birth
- Largest sinus
- Found in body of maxilla
- Has 3 recesses:
 - Apex of maxillary sinus extends into zygomatic bone.
 - Base forms the inferior part of lateral wall of nasal cavity
 - Roof formed by the floor of the orbit.
 - Floor formed by the alveolar part of maxilla.
 - Roots of maxillary teeth (2 molars) produce conical elevations in the floor of sinus

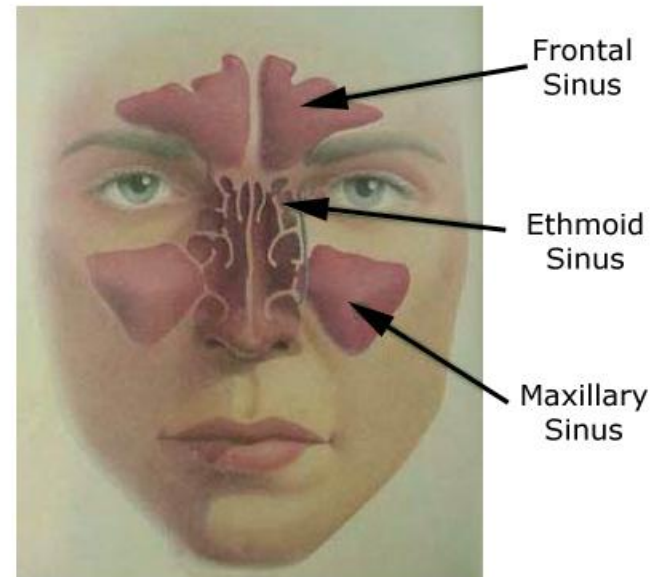


Development of sinuses

- Aerated at birth
 - **Maxillary sinuses**
- Age 6-7
 - **Frontal /sphenoidal sinuses**
- Puberty- approx 17-18 yrs
 - **Ethmoid**



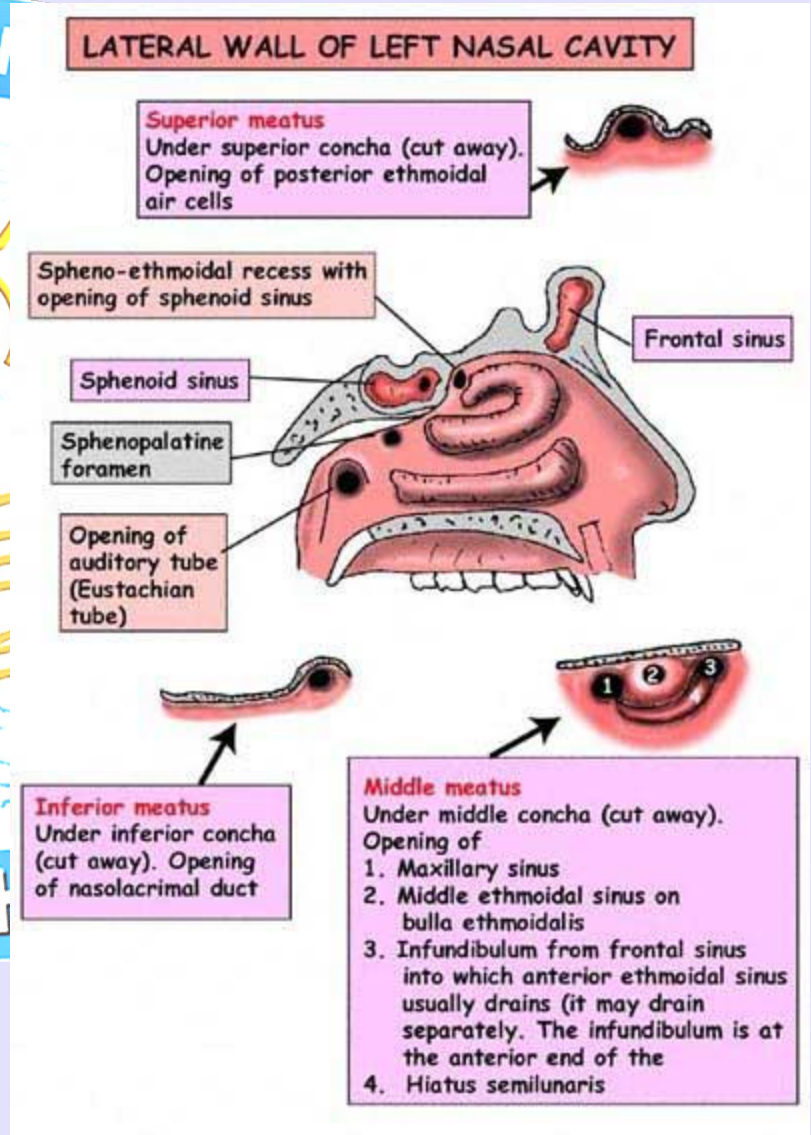
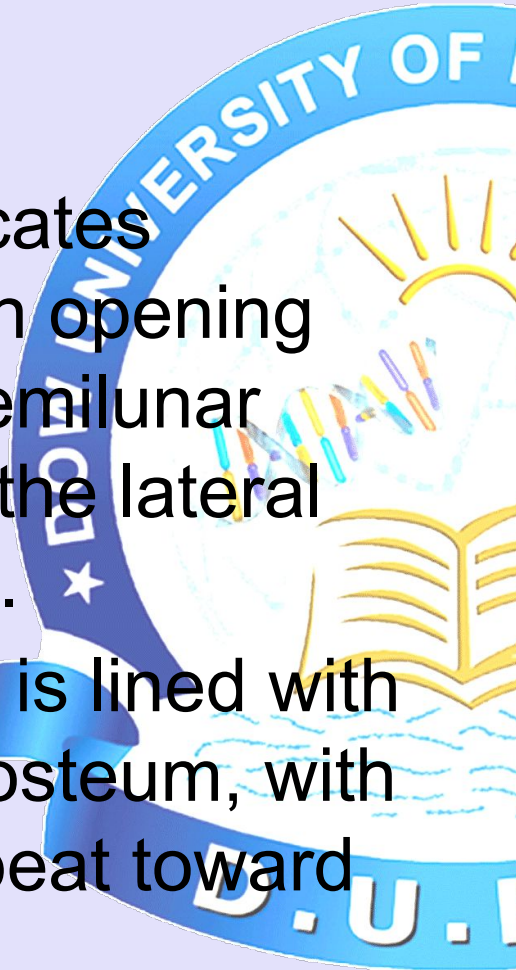
Pediatric Sinus Anatomy



Adult Sinus Anatomy
(Sphenoid Sinus not Shown)

Ostium of maxillary sinus

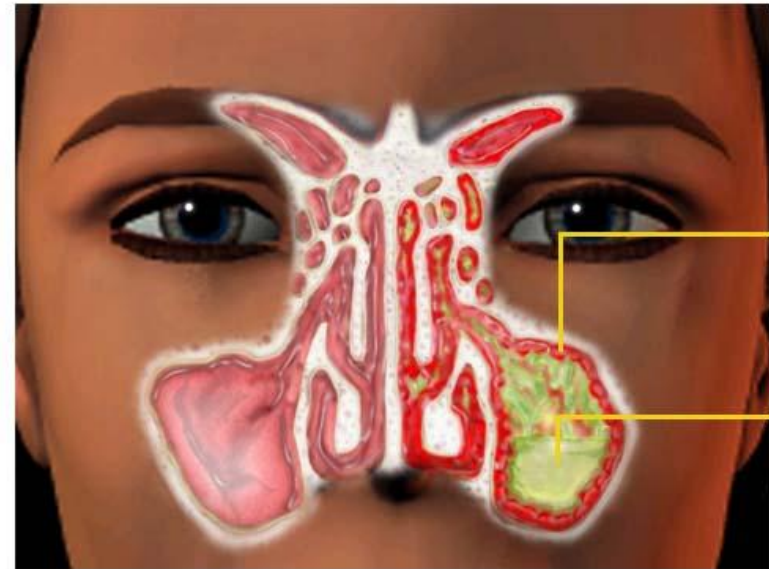
- The sinus communicates through an opening into the semilunar hiatus on the lateral nasal wall.
- The sinus is lined with mucoperiosteum, with cilia that beat toward the ostia



Maxillary sinusitis

- Inflammation of maxillary sinus
- Maxillary sinusitis is common due to the close anatomical relation of the frontal sinus, anterior ethmoidal air sinus and the maxillary teeth, allowing for easy spread of infection.
- Drainage orifice lies near the roof of the sinus, and so the maxillary sinus does not drain at all well, and infection develops more easily.
- Treatment
- Broad spectrum antibiotic

Inflammation of infected left maxillary sinus



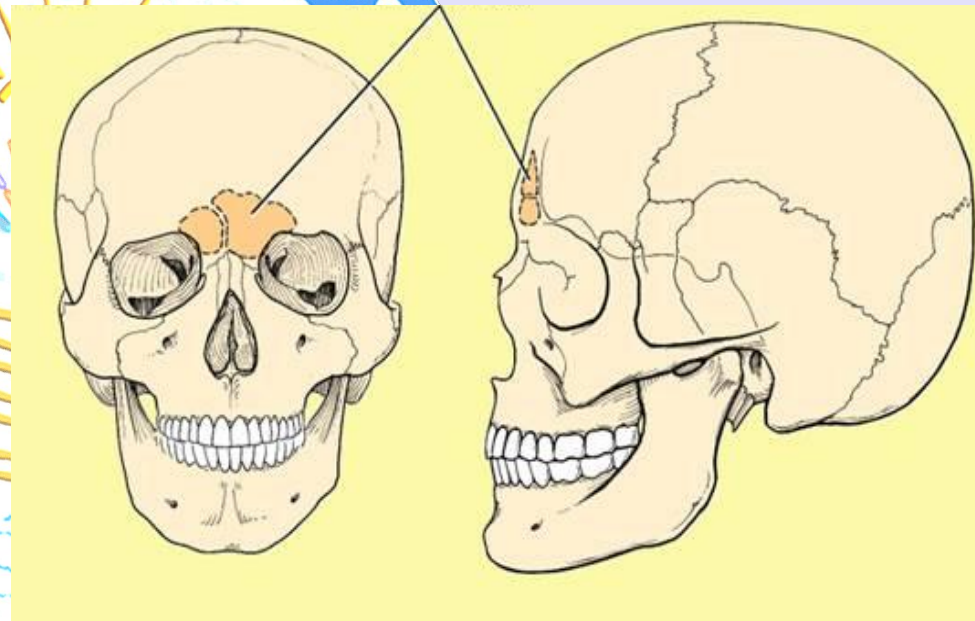
note the red inflamed mucosal tissue as a result of the infection.

Trapped infection begins to exert pressure on the maxillary sinus.

When the ostium narrows, the flow of mucus out of the sinus is restricted thus allowing mucus to back up in the sinus. This pressure from the trapped mucus causes sinus pain.

Frontal sinus

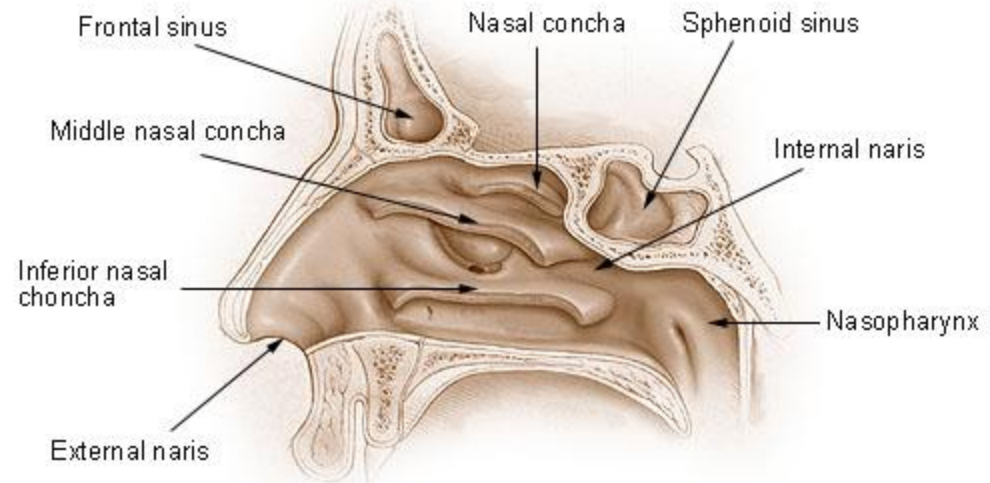
- Second largest sinuses
- Number varies
- Situated behind the superciliary arches,
- Absent at birth, reaching their full size after puberty.
- Frontal sinuses are rarely symmetrical and the septum between them frequently deviates to one or other side of the middle line



Opening of frontal sinus

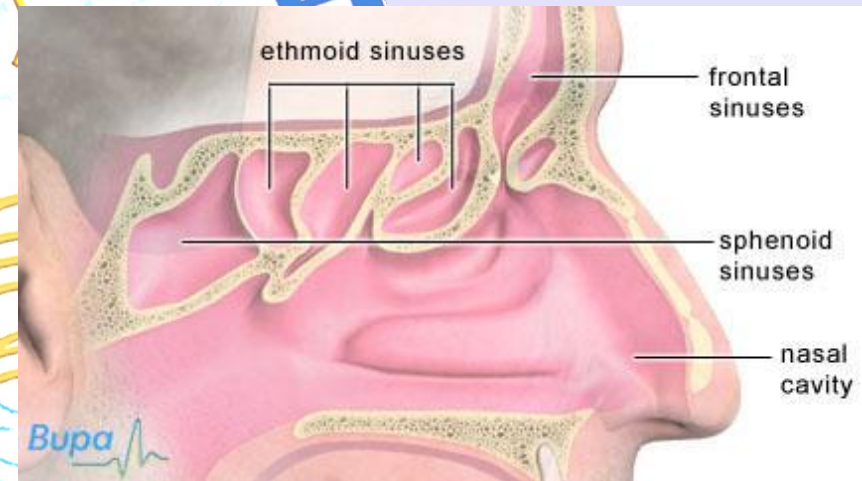
- Opens into the anterior part of the corresponding middle meatus of the nose through the frontonasal duct.
- Each frontal sinus opens into the anterior part of the middle meatus of the nose through the frontonasal duct.
- Which opens into the hiatus semilunaris
- The **hiatus semilunaris** is a halfmoon-shaped groove in the lateral wall of the nasal cavity.

Nose and Nasal Cavities



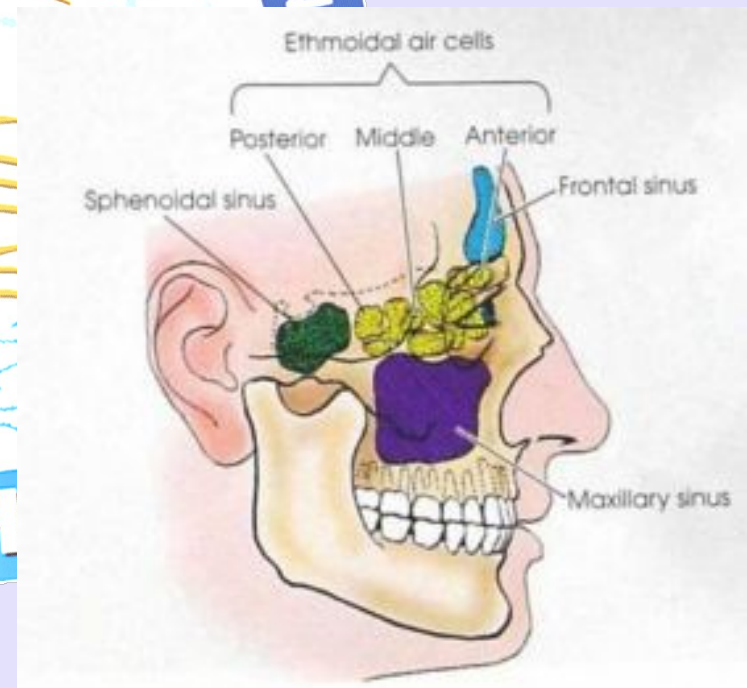
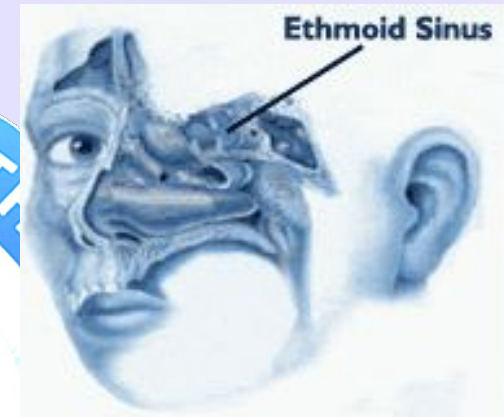
Ethmoid sinuses

- Collective name for the **ethmoidal air cells**.
- Each ethmoid sinus is an air-space enclosed within the ethmoid bone. The ethmoidal air cells consist of numerous thin-walled cavities situated in the ethmoidal labyrinth and completed by the frontal, maxilla, lacrimal, sphenoidal, and palatine bones.
- They lie between the orbits, above the nasal cavity and are separated from these cavities by thin bony laminae.



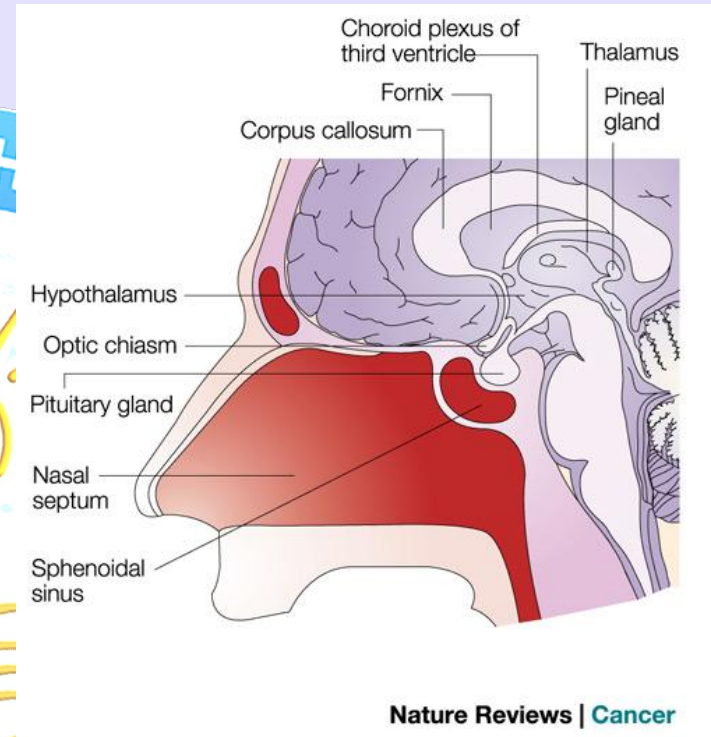
Ethmoid sinuses opening

- Anterior, middle, and posterior ethmoid sinuses.
- **Posterior** drains into the superior meatus under cover of the superior nasal concha; sometimes one or more opens into the sphenoidal sinus.
- **Middle** drains into the middle meatus of the nose on or above the bulla ethmoidalis.
- **Anterior** drains into the middle meatus of the nose by way of the infundibulum.



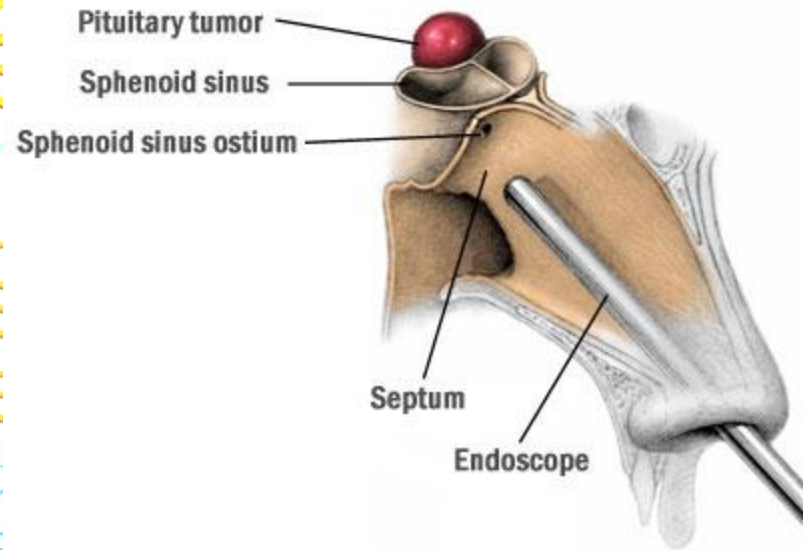
Sphenoid sinus

- They are present as minute cavities at birth, but their main development takes place after puberty.
- Present within the body of sphenoid.
- They are rarely symmetrical due to the lateral displacement of the intervening septum.
- When exceptionally large they may extend into the basilar part of the occipital bone.
- Pituitary gland can be approached through nose bypassing the sphenoid sinus.



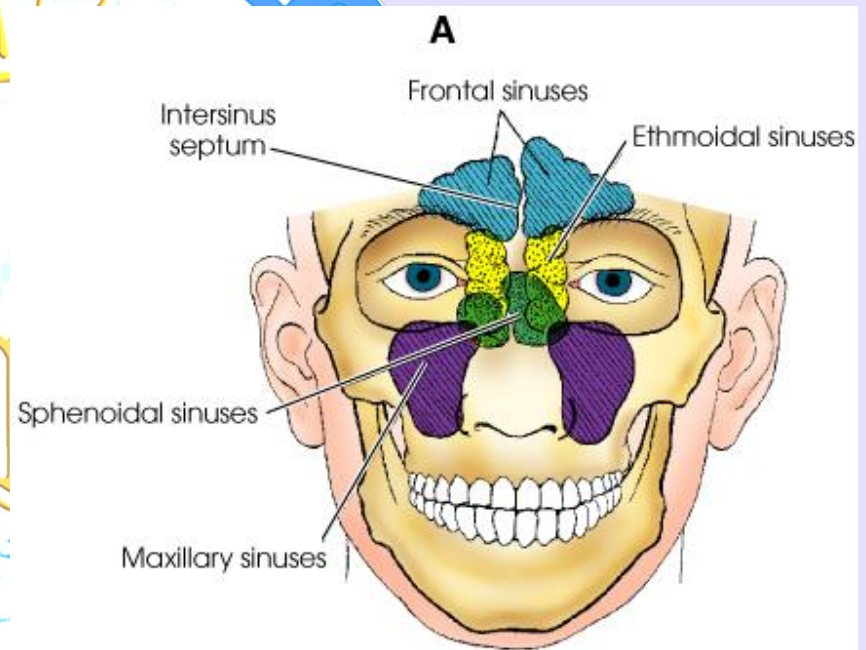
Opening of sphenoid sinus

- Each sinus opens into the roof of the nasal cavity via apertures on the posterior wall of the sphenoethmoidal recess.
- A potential complication of sphenoid sinusitis is cavernous sinus thrombosis.



Function of paranasal sinuses

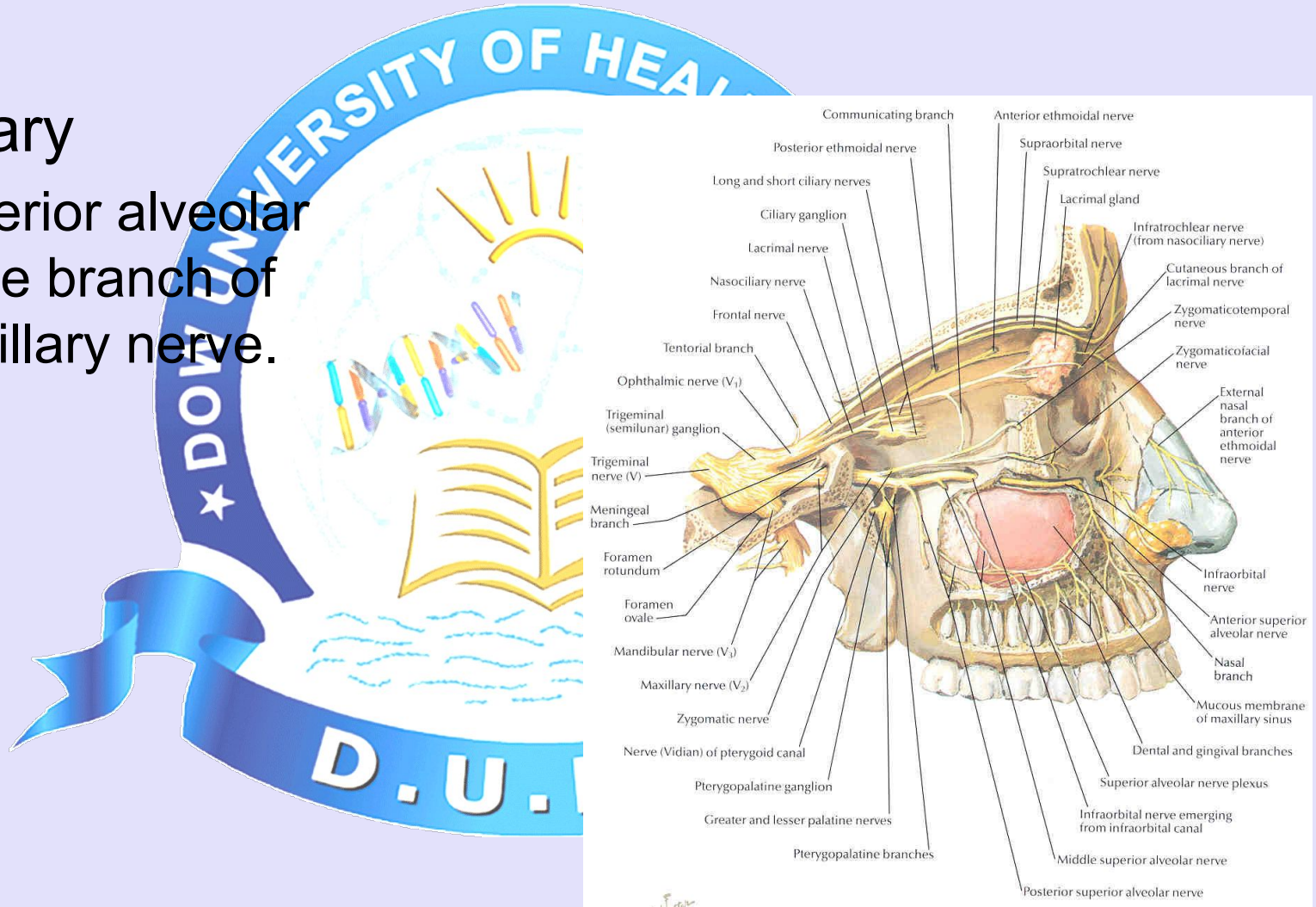
- Resonating chamber for voice
- Decrease weight of skull
- Warm & moisten air
- Shock absorbers
- Immune system



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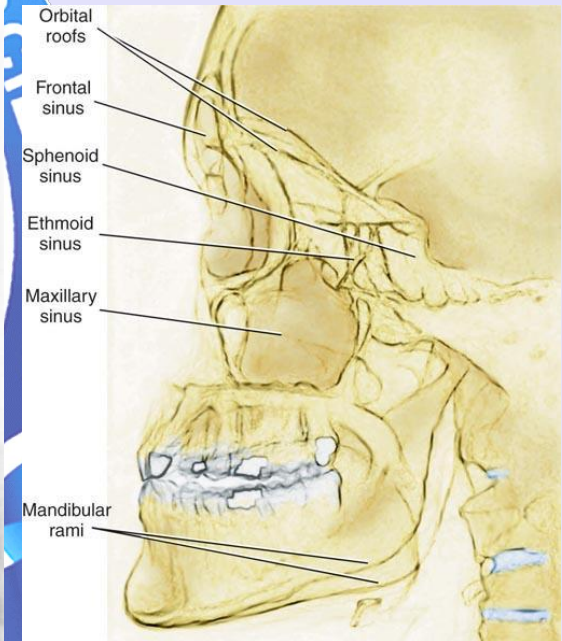
Innervation

- Maxillary
 - Superior alveolar nerve branch of maxillary nerve.

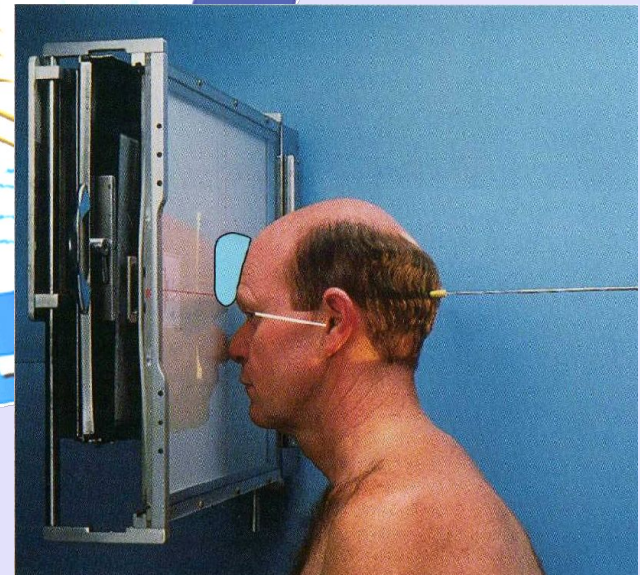
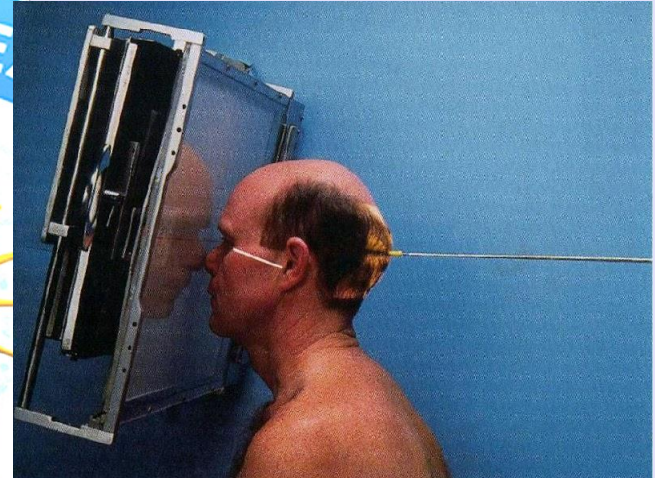
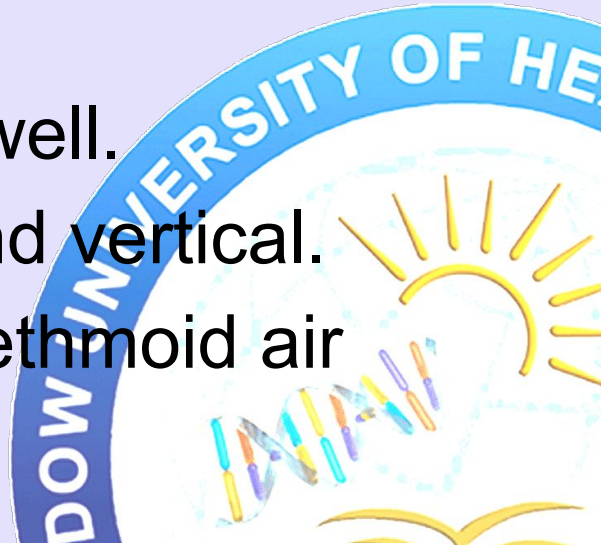


Radiology

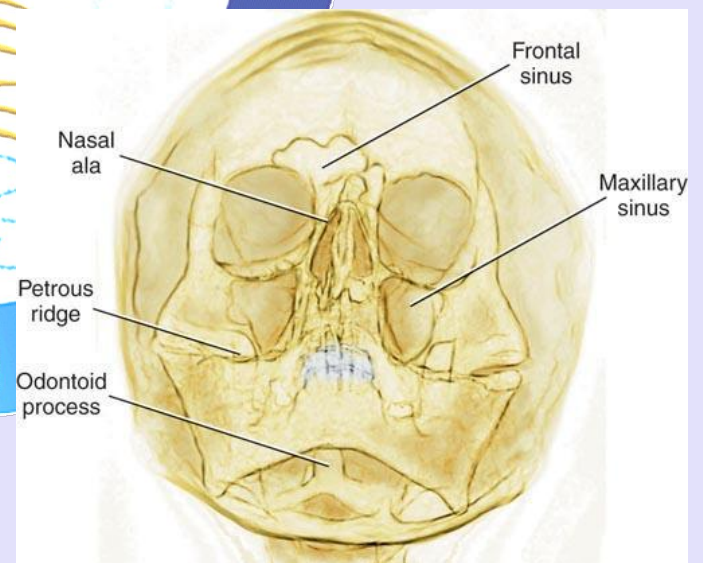
- Lateral Radiograph
- All 4 sinuses
 - Sphenoid of primary interest



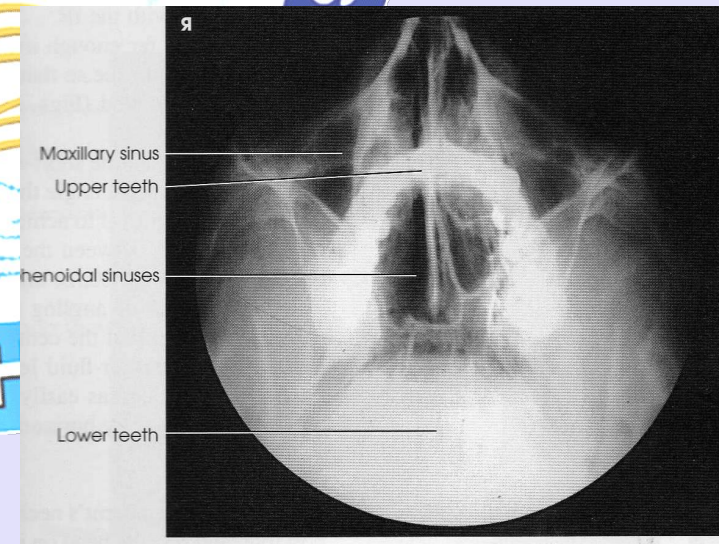
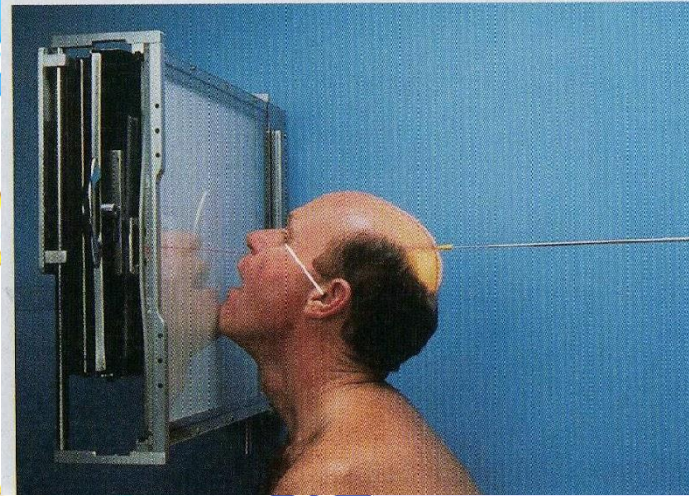
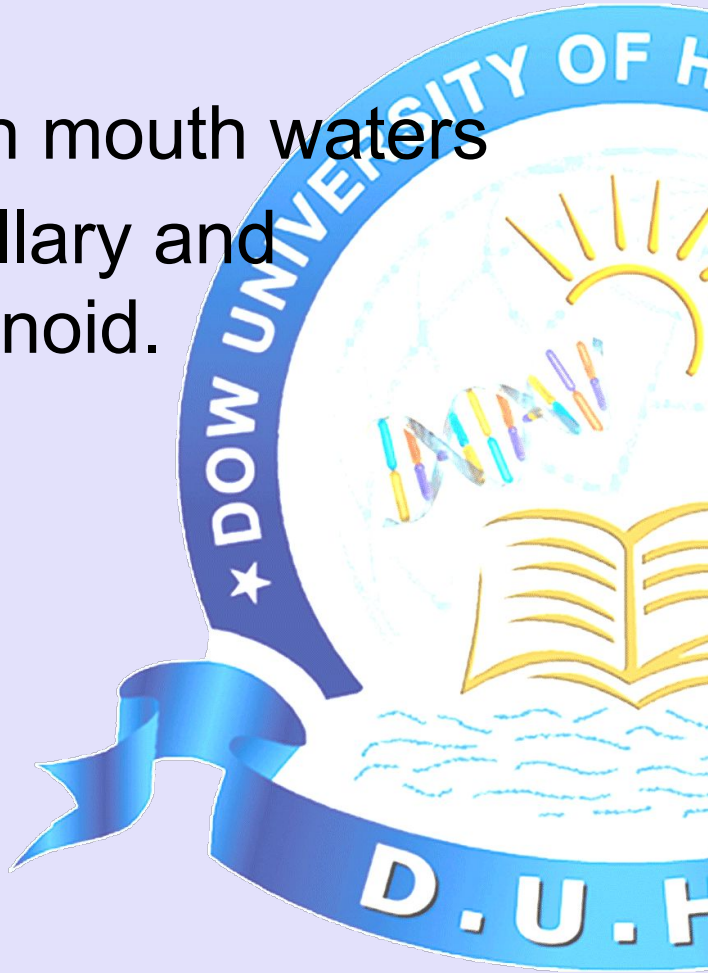
- (PA) caldwell.
- Angled and vertical.
- Frontal & ethmoid air cells



- Waters view.
- For maxillary sinus



- Open mouth waters
- Maxillary and sphenoid.



Identify???

